

MACHINE LEARNING METHODOLOGIES

SUMMARY III (Version 1)

Name: AKINSANYA OKIKIOLA MOSES

Student ID: VEPH/50B/AI0005.

This summary contains what I understand from the first resource on a Machine Learning Methodology; **FEATURE SELECTION**.

In a machine learning dataset, a **feature** generally refers to an input variable or column that is used by a model to make predictions. The target variable, on the other hand, is what the model is trying to predict.

Feature Selection is the process of identifying and selecting the most useful features to include when training a machine learning model.

The goal is not necessarily to use every available column. Instead, we want to determine which features provide useful information for predicting the target and remove features that contribute little or no useful information.

For example, if a dataset contains 30 columns but only 12 provide meaningful predictive information, feature selection can help us focus the model on those useful features.

Why Do We Perform Feature Selection?

Feature selection is important for several reasons.

Simpler Models

If two models provide similar performance, a **simpler model with fewer relevant features is generally preferable**.

Using fewer features can make a model easier to understand, interpret, and maintain.

Faster Training

Fewer features mean less information for the algorithm to process.

This can reduce:

- Training time
- Computational requirements
- Memory usage

This becomes particularly important when working with large datasets containing hundreds or thousands of features.

Potentially Better Model Performance

Not every feature contains useful predictive information.

Some features may introduce noise or irrelevant information. Removing such features can sometimes improve a model's ability to generalize to unseen data.

Feature selection can therefore help reduce unnecessary complexity and potentially improve predictive performance.

When Should Feature Selection Be Performed?

Feature selection should generally be considered **before hyperparameter tuning and model selection**.

The reason is that the features used to train the model can affect which model and hyperparameters perform best.

A simplified machine learning workflow can therefore look like:

Data Preparation → Feature Selection → Model Selection → Hyperparameter Tuning → Model Evaluation

However, the exact workflow can vary depending on the project and feature-selection technique being used.

An important practical consideration is **data leakage**: feature selection should be performed using training data when the selection process uses information from the target. The test set should remain unseen until final evaluation.

Main Feature Selection Methods

The resource introduced four important feature-selection approaches:

- Variance Threshold
- K-Best Features
- Recursive Feature Elimination (RFE)
- Boruta

These methods approach feature selection differently.

Variance Threshold

Variance measures how much the values of a feature vary around their mean.

A feature with very little variation may provide very little useful information.

The extreme example is a feature with a **variance of 0**.

This means every observation has exactly the same value.

For example, if every record has:

Country = Nigeria

then the feature contains no variation within that dataset.

Because every observation has the same value, the feature cannot help the model distinguish between observations.

Variance Threshold Method

The **Variance Threshold** method removes features whose variance falls below a specified threshold.

The simplest case is removing constant features with variance equal to zero.

Key idea

Low variance → potentially little useful information

However, low variance does not automatically mean a feature is useless. A feature with low variance could still be predictive, depending on the problem.

Therefore, the threshold should be chosen carefully.

K-Best Features

The **K-Best Features** approach is a **filter method**.

Instead of training a model to determine which features are important, it uses a statistical measure to evaluate the relationship between individual features and the target.

The method then selects the **top K features** according to the chosen scoring method.

For example, if you specify $K = 10$, the method selects the ten features that receive the highest scores according to the chosen statistical test.

Why is it called a Filter Method?

It is called a filter method because the features are filtered **before the model is trained**, based on statistical characteristics rather than model-specific importance.

Choosing the Statistical Test

The appropriate statistical test depends on the type of the **feature** and the **target**.

The resource identifies the following combinations:

Categorical Feature → Categorical Target

Methods such as the **Chi-square test** can be used to examine the relationship between categorical variables.

Categorical Feature → Numerical Target

A **t-test** may be used in appropriate two-group situations to compare numerical outcomes across categorical groups.

Numerical Feature → Categorical Target

Statistical tests such as a **t-test** can be relevant in appropriate situations, particularly when comparing a numerical variable across two groups.

For more than two groups, other methods may be more appropriate.

Numerical Feature → Numerical Target

Measures such as:

- **Pearson correlation**
- **Spearman's rank correlation**

can be used to assess relationships between numerical variables.

Pearson correlation primarily measures the strength of a **linear relationship**, while Spearman correlation is based on the ranks of the observations and can capture **monotonic relationships** that are not necessarily linear.

Important Point

The statistical method should match the type of data and the assumptions of the method. "K-Best" describes the selection approach; the specific scoring function determines how features are evaluated.

Recursive Feature Elimination (RFE)

Recursive Feature Elimination (RFE) is a **wrapper method**.

Unlike a filter method, RFE involves using a machine learning model to determine which features are important.

The basic idea is:

Train → Rank Features → Remove the Least Important → Retrain → Repeat

RFE starts with the available features and repeatedly removes the least important ones until the desired number of features remains.

Model Feature Importance

RFE needs a model that can provide some measure of **feature importance or coefficient importance**.

For example, certain tree-based models can calculate feature importance.

The model is therefore actively involved in deciding which features should remain.

Why is RFE Called a Wrapper Method?

It is called a wrapper method because the feature-selection process **wraps around the machine learning model**.

The model is repeatedly trained as features are removed.

Advantage

RFE can identify features based on their usefulness to a particular model.

Limitation

Because the model may need to be trained repeatedly, RFE can be more computationally expensive than simpler filter methods.

Boruta

Boruta is another feature-selection algorithm that uses a model-based approach.

One of its main ideas is to make the selection process less dependent on manually choosing an arbitrary importance threshold.

The resource describes this idea as "**taking the human out**" because Boruta establishes its comparison threshold from the data rather than requiring the user to manually decide what importance score is sufficient.

Boruta commonly works with a tree-based model to evaluate feature importance.

How Boruta Works

The process can be understood through the idea of **shadow features**.

Create Shadow Features

Boruta creates copies of the original features and randomly shuffles their values.

These are called **shadow features**.

Because their values have been randomly shuffled, they should no longer retain the meaningful relationship between the original feature and the target.

Calculate Feature Importance

The importance of the original features and shadow features is then calculated using the underlying model.

Establish a Benchmark

The highest importance score among the shadow features becomes the benchmark or threshold for comparison.

This gives Boruta a reference point representing the level of importance that could potentially be achieved by random information.

Compare Original Features

The importance of each original feature is compared with the shadow-feature benchmark.

If an original feature performs better than the benchmark, it provides evidence that the feature contains useful predictive information.

Features that do not demonstrate sufficient importance can be rejected.

Key Idea

Boruta essentially asks:

Is this real feature more informative than what we could obtain from a randomized version of the data?

This makes the method useful when the goal is to identify genuinely relevant features rather than simply selecting a fixed number of the highest-ranked features.

Comparing the Methods

The four methods differ mainly in **how they decide whether a feature is useful**.

Variance Threshold

Looks at how much a feature varies. It is useful for removing constant or very low-variance features.

K-Best

Uses statistical tests to rank features according to their relationship with the target and selects the top K.

RFE

Uses a machine learning model to repeatedly remove less important features.

Boruta

Uses model-based feature importance and compares real features against randomized shadow features.

A simpler way to remember all the methods is to think of them as asking different questions:

Variance Threshold:

"Does this feature vary enough to potentially be useful?"

K-Best:

"Which features have the strongest statistical relationship with the target?"

RFE:

"Which features does my chosen model find most useful?"

Boruta:

"Is this feature more useful than what could be achieved by random information?"

Important Considerations

Feature selection is not simply about removing as many columns as possible.

A feature should be removed because there is a reasonable basis for considering it unnecessary, not simply because it looks inconvenient.

It is also important to distinguish **feature selection** from **feature engineering**.

Feature selection chooses from features that already exist.

Feature engineering creates, transforms, or combines features to produce potentially more useful inputs.

Another important consideration is that the usefulness of a feature can depend on the model being used. A feature that is useful to one model may not provide the same benefit to another.

Finally, feature selection should be performed carefully to avoid **data leakage**, especially when the selection method uses the target variable. Information from the test set should not influence the feature-selection process.

Overall Takeaway

Feature Selection is an important part of the machine learning methodology because having more features does not automatically mean having a better model.

The objective is to identify a **useful and manageable set of features** that provides the information the model needs to make good predictions.

The key methods covered are:

Variance Threshold → removes features with insufficient variation.

K-Best → statistically evaluates features and keeps the top K.

RFE → repeatedly uses a model to eliminate less important features.

Boruta → compares real features against randomized shadow features to identify genuinely relevant predictors.

The broader principle is simple:

Good feature selection aims to keep useful information while reducing unnecessary noise and complexity.